

Density-Dependent Coherence Integration: Detailed Derivation of Conscious Coherence Term in Effective Density - v12.1

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Abstract

This paper presents the corrected derivation of the conscious coherence contribution to effective energy density $\rho_{\text{eff}}(a)$ in Horndeski-based EFT of Dark Energy. Versions v1-v11 coupled ρ_{psi} via $\rho_{\text{psi}} * f_L$ or M_B , which is inconsistent with Bellini & Sawicki. v12.1 corrects to additive coupling. $\Phi_t = 0.6 * \text{EEG_coh} + 0.4 * \text{HRV_coh}$ in $[0,1]$, $\rho_{\text{psi}} = \Xi * \epsilon_0 * \Phi_t$ [kg/m³], $\Xi * \epsilon_0 = 1.0e-26$ kg/m³ reference. Background corrected: $R(a) = (1+r)/(1+2r)$, $r(a) = G_4\phi/(X G_3X - G_4\phi) = 1/(1+\epsilon(a))$, $\epsilon(a) = X G_3X/G_4\phi - 2$. $\rho_m(a) = \rho_{m0} a^{-3}$ replaces ρ_L in background. Model is separately falsifiable by Euclid/DESI (h_i class 0.03% Planck) and EEG/HRV correlation.

1. Introduction

Time Attraction v1-v11 introduced $f_L = \exp(-\gamma_L \rho_L)$, $\gamma_L = G_N L^2 / c^2$, lab prediction $\Delta f/f > 1e-18$ for $\Delta \rho = 1e4$ kg/m³. Previous coupling $\rho_{\text{psi}} * f_L(R)$ used $R_{\text{old}} = (r+1)/(r+2)$, causing 25-36% systematic error. This work implements correction following Bellini & Sawicki Eq.2.18 and Jordi Sec V, decoupling f_L from background EFT functions $\alpha_M(a)$, $\alpha_B(a)$.

2. Coherence Metrics and Φ_t

EEG_coh in $[0,1]$: alpha band 8-12 Hz inter-hemispheric coherence $|S_{xy}|/\sqrt{S_{xx}S_{yy}}$ or PLV, 5 min, 19 channels. 0=incoherent, 1=fully coherent.

HRV_coh in $[0,1]$: heart coherence ratio peak power/total 0.04-0.26 Hz or normalized RMSSD, 5 min EKG/PPG. 0=chaotic, 1=sinusoidal 0.1 Hz.

Weights from cortical dominance 60/40, normalized $w_{\text{EEG}} + w_{\text{HRV}} = 1$.

$$\Phi_t = 0.6 * \text{EEG_coh} + 0.4 * \text{HRV_coh} \text{ in } [0,1] \quad (1)$$

3. Physical Density ρ_{psi}

ϵ_0 is normalization, Ξ coupling scale. Reference $\Xi * \epsilon_0 = 1.0e-26$ kg/m³ sets order of magnitude.

$$\rho_{\text{psi}} = \Xi * \epsilon_0 * \Phi_t \text{ [kg/m}^3] \quad (2)$$

Example: EEG_coh=0.70, HRV_coh=0.80 => $\Phi_t=0.74$ => $\rho_{\text{psi}}=7.4e-27$ kg/m³.

4. Background Correction and $R(r)$

Correction: ρ_L replaced by $\rho_m(a) = \rho_{m0} a^{-3}$, $a = 1/(1+z)$. $r(a)$ independent of f_L : $r = \alpha_0 f_L / [\alpha_0 f_L (1+\epsilon)] = 1/(1+\epsilon)$. $M^2 = 2 G_4$, $\alpha_M = 2 \phi \dot{G}_4\phi / (H M^2)$, $\alpha_B = 2 \phi \dot{X G_3X - G_4\phi}$.

Bellini & Sawicki Eq.2.18 corrected form:

$$R(a) = (1+r(a))/(1+2r(a)), \quad dR/dr = -1/(1+2r)^2 < 0 \quad (3)$$

$$r(a) = (1-R)/(2R-1) = G_4\phi/(X G_3X - G_4\phi) = 1/(1+\epsilon(a)) \quad (4)$$

$$\epsilon(a) = X G_3X/G_4\phi - 2 \quad (5)$$

$$\text{Locus: } \epsilon=0 \iff X G_3X=2 G_4\phi \iff r=1 \iff R=2/3 \quad (6)$$

5. Integration into $\rho_{\text{eff}}(a)$

Previous error: multiplicative $\rho_{\text{psi}} * f_L$ invalid because f_L not background EFT function. Corrected additive form:

$$\rho_{\text{eff}}(a) = \rho_{b0} * M_{\text{Rho}}(R(a)) + \rho_{DM0} * \rho(R(a)) + \rho_{B0} * M_B/R(a) + \rho_{GR} * M_B/R(a) + \rho_{E0} * M_B/R(a) + \rho_{\text{psi}}(\Phi_t) \quad (7)$$

All $M_i(R(a))$ depend on background $R(a)$ only. $\Phi_t(a)$ is binned temporally and correlated with $a(t)$ without multiplicative coupling. This restores EFT consistency.

6. Full Formula Collection v12.1

$$R = (1+r)/(1+2r) \text{ [Bellini Eq.2.18]}$$

$$r = G4_{\Phi}/(X G3X - G4_{\Phi}) = 1/(1+\epsilon)$$

$$\epsilon = X G3X/G4_{\Phi} - 2$$

$$M^2 = 2 G4, \alpha_M = 2 \dot{\Phi} G4_{\Phi}/(H M^2)$$

$$\alpha_B = 2 \dot{\Phi} [X G3X - G4_{\Phi}]$$

$$\rho_m(a) = \rho_{m0} a^{-3}, a = 1/(1+z)$$

$$\Phi_t = 0.6 \cdot \text{EEG}_{\text{coh}} + 0.4 \cdot \text{HRV}_{\text{coh}}$$

$$\rho_{\psi} = \Xi \epsilon_0 \Phi_t$$

$$\rho_{\text{eff}} = \sum \rho_0 M_i(R) + \rho_{\psi}$$

$$f_L = \exp(-\gamma_L \rho_L), \gamma_L = G_N L^2/c^2$$

Mennyiség	Kifejezés	Példa
R	$(1+r)/(1+2r)$	$r=0.5 \rightarrow 0.75$
r	$1/(1+\epsilon)$	$\epsilon=0 \rightarrow 1$
Φ_t	$0.6 \cdot \text{EEG} + 0.4 \cdot \text{HRV}$	0.74
ρ_{ψ}	$\Xi \epsilon_0 \Phi_t$	$7.4e-27 \text{ kg/m}^3$
ρ_{eff}	$\sum M_i(R) + \rho_{\psi}$	additív

7. Falsifiability and Conclusion

Cosmology: model fails if $R(a) \neq (1+r)/(1+2r)$ or h_i class not 0.03% Planck recovery with $r=1..1.3$ in Euclid/DESI 4x3 tomographic bins.

Consciousness: fails if Φ_t shows no correlation with $a(t)$ or $\rho_{\psi}=0$, i.e. EEG_coh and HRV_coh random.

Laboratory: $f_L = \exp(-\gamma_L \rho_L)$, $\Delta \rho = 1e4 \text{ kg/m}^3$ must show $\Delta f/f > 1e-18$ (lead-vacuum 1.44 ns/s, tungsten-vacuum 2.45 ns/s) with atomic clocks.

Conclusion: v12.1 provides corrected, EFT-consistent, additive coupling $\rho_{\text{eff}}(a) = \sum \rho_0 M_i(R(a)) + \rho_{\psi}(\Phi_t)$, with $R=(1+r)/(1+2r)$, $r=1/(1+\epsilon)$, $\epsilon=X G3X/G4_{\Phi} - 2$, $\rho_m(a)$ background, and separately testable predictions.

References

[1] Bellini & Sawicki 2014 Eq.2.18

[2] Jordi, Sec V

[3] Planck 2018

[4] h_i class, Euclid, DESI

[5] EEG coherence $|S_{xy}|/\sqrt{S_{xx}S_{yy}}$, HRV coherence 0.04-0.26 Hz